MEDopathies Alliance

MED12 · MED13 · MED13L

Virtual Scientific Symposium

Friday, October 9, 2026 · 10:00–12:00 & 13:00–16:00 U.S. Pacific · Zoom

A single day bringing together researchers, clinicians, biotech and pharma, and the patient communities across the three Mediator-complex neurodevelopmental disorders.

Save the date · registration opens soon

Free and open. Sessions are recorded; proceedings will be posted publicly after the event.

WHY ONE ALLIANCE

MED12, MED13, and MED13L are regulatory subunits of a single molecular machine: the CDK8 kinase module of the Mediator complex, the switchboard every cell uses to turn genes on and off during brain development. The three disorders share structure, overlapping clinical features, and a set of open scientific questions. This symposium is the Medopathies Alliance's first public program — one module, three genes, one community.

Morning · Mechanism and Models 10:00–12:00 PT

10:00–10:10

Welcome: the case for the Medopathies Alliance

A brief joint opening from the leaders of the three foundations: Amy Krystosik, PhD, MPH (MED13), Katie Barry Boychuck (MED13L), and Kelly Sarber (MED12).

10:10–10:55

OPENING KEYNOTE

The Mediator complex and the CDK8 kinase module

Cryo-EM of the human Mediator kinase module and the structural target landscape, with emphasis on the structural distinction between MED13 and MED13L. Followed by audience Q&A.

Dr. Kuang-Lei Tsai, UTHealth Houston · introduced and moderated by **Dr. Dylan Taatjes**, University of Colorado Boulder

11:00–12:00

PANEL

Model systems across the Mediator spectrum

One module, three genes, one open question: what each model system can and cannot tell us, where the three disorders converge, and where they do not.

With **Zexuan Li**, UC Santa Cruz (mouse models of MED13 and cortical development) and **Dr. Maya Arce**, Arc Institute (MED12 as a transcriptional coordinator; CRISPR screening in primary human cells). Additional panelists to be announced. Moderated by **Dr. Dylan Taatjes**.

MIDDAY BREAK · 12:00–13:00 PT

Afternoon · From Mechanism to Medicine 13:00–16:00 PT

13:00–13:35

Natural history, phenotype, and cohort readiness

What we know about the course of these disorders, and how registries and biorepositories make the cohort trial-ready.

With **Dr. Carla Marini**, Salesi Children's Hospital, Ancona, and **Lindsay Wieczorek**, COMBINEDBrain, on the biorepository. Additional speakers to be announced. Moderated by **Terry Jo Bichell, PhD**, COMBINEDBrain.

13:35–14:25

PANEL

Mechanism before modality

MED13L includes well-documented dosage loss and also missense variants of unresolved mechanism. MED13 carries both truncating and missense variants. The classic X-linked MED12 syndromes are missense, and at least one appears to act by failing to switch the CDK8 kinase on rather than by reducing how much protein is there. Within one module — and sometimes within one gene — different variants are not breaking the same thing. Until we can assign a mechanism class variant by variant, we cannot honestly tell a family which therapeutic path is theirs.

With **Dr. Ryan L. Boudreau**, University of Iowa (RNA-based control of gene dosage: antisense oligonucleotides and AAV-delivered gene modulation) and **Dr. Nadav Ahituv**, UCSF (CRISPR activation and cis-regulation therapy). Additional panelists to be announced. Moderated by **Ricardo N. Ramirez, PhD**, MED13L Foundation.

14:25–14:55

KEYNOTE

Federal rare disease research infrastructure

How ultra-rare research programs get built and funded. Followed by Q&A.

Dr. P.J. Brooks, NIH / NCATS

BREAK · 14:55–15:05 PT

15:05–15:55

FAMILY PANEL

Living with a disorder of this module

A conversation, not prepared remarks, among parents from all three communities: two families each from MED12, MED13, and MED13L, invited by their own foundation. Moderated by **Jean Chia, MD**, Stanford Medicine Children's Health. This session is recorded for the communities but excluded from the open license, and every panelist retains the right to review and withdraw.

15:55–16:00

Closing: the Alliance roadmap

What comes next: the Medopathies Research Workshop in Denver (December 2026, alongside the AES Annual Meeting) and the Meeting of the MEDs (June 2027). All three foundations.

Featured speakers

The full speaker list will be published with the final program.



Kuang-Lei Tsai, PhD OPENING KEYNOTE

Associate Professor, Biochemistry & Molecular Biology, McGovern Medical School, UTHealth Houston

Internationally recognized for pioneering structural work on the Mediator complex and its CDK8 kinase module, including studies that uncovered the critical role of the MED13 subunit in regulating Mediator activity. His laboratory uses cryo-electron microscopy with biochemical and biophysical approaches to understand how transcriptional dysregulation contributes to human disease.



Dylan Taatjes, PhD MODERATOR

Professor, Biochemistry, University of Colorado Boulder

The Taatjes lab studies the molecular mechanisms that control human gene expression, focusing on the Mediator complex within the transcription pre-initiation machinery, and how transcription and signaling pathways are altered in development and disease.



Carla Marini, MD, PhD

Director, Pediatric Neuropsychiatry Unit, Salesi Children's Hospital, Ancona, Italy

A pediatric neurologist and internationally recognized expert in pediatric epilepsy, neurogenetics, and developmental epileptic encephalopathies, with more than 300 peer-reviewed publications contributing to the discovery of disease-causing genes and precision medicine in neurological disorders.



Maya Arce, PhD

Science Fellow, Arc Institute

Leads a Arc Institute group using large genomic datasets and CRISPR screening to understand how genetic variation affects immune cell function. During her PhD at UCSF in the Alex Marson Lab, her first-author Nature study showed how MED12 acts as a central coordinator of the dynamic gene circuits governing T cell rest and activation, work that expanded the field's understanding of the Mediator complex in specific cell types.



Nadav Ahituv, PhD

Professor, Bioengineering & Therapeutic Sciences, UCSF

His laboratory links gene regulatory elements and their variation to human disease, and pioneered cis-regulation therapy: using regulatory elements as therapeutic targets for haploinsufficient disorders. His team recently demonstrated CRISPR activation rescue of SCN2A haploinsufficiency in mouse and in human stem-cell-derived neurons (Nature 2025;646:983).



Terry Jo Bichell, MPH, PhD MODERATOR

Founder & Chief Executive Officer, COMBINEDBrain

A neuroscientist and former public health nurse-midwife who launched COMBINEDBrain in 2019, a pre-competitive consortium of patient advocacy organizations working to identify outcome measures and biomarkers for rare genetic neurodevelopmental disorders. She earned her PhD in neuroscience from Vanderbilt, served on the National Academies committee on drug evaluation for rare diseases, and came to this work as the parent of a son with Angelman syndrome.



Lindsay Wieczorek, PhD

COMBINEDBrain · on the biorepository

A biomedical scientist and rare disease advocate with over 20 years of experience across immunology, infectious disease, and neurodevelopmental research. Previously Scientific Director of the Syngap Research Fund, where she led real-world data initiatives for drug repurposing, and co-director of the Humoral Immunology Section at the Walter Reed Army Institute of Research. She is the parent of a child with SYNGAP1-related disorder.



Ricardo N. Ramirez, PhD MODERATOR

Chief Scientific Officer, MED13L Foundation

A genome and data science leader with deep expertise in epigenetics, computational biology, immunology, and gene-editing technologies. He earned his PhD at UC Irvine, contributing to the ENCODE and STATegra consortia, completed a postdoctoral fellowship at Harvard Medical School with the ImmGen consortium, and directed genomic specificity and safety strategy for next-generation epigenetic medicines as a biotech data science lead. As CSO of the MED13L Foundation, he leads investigational research and preclinical strategy toward viable therapies for MED13L syndrome.



Philip J. (P.J.) Brooks, PhD KEYNOTE

Acting Director, Division of Rare Diseases Research Innovation, NCATS, National Institutes of Health

Leads NCATS' rare disease research division and represents NCATS in the NIH-wide Gene Therapy Working Group and the International Rare Diseases Research Consortium. Co-coordinator of the NIH Common Fund program on Somatic Cell Genome Editing and co-chair of the Bespoke Gene Therapy Consortium; recipient of the 2022 Sonia Skarlatos Public Service Award from the American Society of Gene & Cell Therapy.



Jean Chia, MD FAMILY PANEL MODERATOR

Clinical Assistant Professor, Pediatrics · Complex Primary Care Clinic, Stanford Medicine Children's Health

A board-certified pediatrician caring for children with medical complexity at Stanford Medicine Children's Health's Complex Primary Care Clinic. Trained at Northwestern University Feinberg School of Medicine with pediatric residency at NYU, she brings a clinician's perspective to conversations between families, researchers, and care teams.



Zexuan Li

PhD student, Molecular, Cell & Developmental Biology, UC Santa Cruz

Studies the molecular mechanisms underlying neurodevelopmental disorders. His first-author study in Communications Biology demonstrated that Med13 regulates cortical neuron radial migration and contralateral projection through the PlxnA4 signaling pathway, providing new mechanistic insight into MED13-associated disorders.

RECORDING AND ACCESS

Open proceedings

Keynotes and scientific sessions are recorded and posted publicly under CC BY 4.0, together with short edited clips of the keynotes and panels.

The family panel is different

Hosted by the Alliance, all rights reserved, and revocable at any time by any panelist. An open license is irrevocable, so we chose the right to withdraw.

No identifiable participant data

Because sessions are posted publicly, no session displays individual-level genotype–phenotype tables, clinical photographs, or other identifiable data. Slides are reviewed in September.

Medopathies Alliance
A program of the MED12, MED13, and MED13L foundations

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